# Orthogonal Rotation of Principal Component Factors in Affine Term Structure Models

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**Abstract**

This paper extends the affine term structure model (ATSM) introduced by Adrian, Crump, and Moench (2013), hereafter ACM. The ACM framework extracts latent pricing factors from a panel of zero-coupon yields via principal component analysis (PCA) and models their dynamics using a vector autoregression (VAR). While effective, this approach lacks an economically interpretable structure for the PCA factors. We address this limitation by introducing a rotation and scaling procedure based on Givens transformations, which imposes economically meaningful restrictions on the factor loadings. This modification preserves the model's ability to capture the cross-sectional shape of the yield curve while enhancing interpretability and enabling the integration of macroeconomic priors. Specifically, by aligning the first factor with the short-term interest rate and imposing sparsity in the loading matrix, the framework facilitates the incorporation of prior beliefs about monetary policy transmission and macro-financial linkages. This approach improves transparency in yield curve dynamics, supports structural consistency with macroeconomic theory, and maintains tractability for estimation and forecasting.

## 1 Orthogonal Rotation of Principal Component Factors in Affine Term Structure Models

This paper presents an extension of the yield curve modeling framework introduced by Adrian, Crump, and Moench (2013), hereafter ACM. The original ACM model estimates the latent factors from a panel of zero-coupon yields using principal component analysis (PCA), and models their dynamics using a vector autoregression (VAR). Although effective in capturing the main sources of variation in the yield curve, the PCA-based factors lack direct economic interpretation due to their arbitrary orientation and scale.

To address this limitation, we propose a modification that imposes an economically motivated structure on the PCA factors through a sequence of Givens rotations and a subsequent scaling procedure. Rotating the PCA-extracted factors allows for a more interpretable factor structure, in which the first factor can be aligned with the short-term risk-free interest rate. This transformation enables the imposition of economically meaningful priors on the unconditional mean of the first factor that can be used within a Bayesian VAR framework.

### 1.1 The ACM Yield Curve Model

The model developed by Adrian, Crump, and Moench (2013) is part of the class of affine term structure models (ATSM), which describe bond yields as affine functions, that is, linear combinations plus a constant, of the underlying $k$ factors. This structure provides analytical tractability while allowing for a flexible representation of the yield curve as described by Duffie and Kan (1996). In this framework, the yield on a zero-coupon bond with maturity $n$ at time $t$ is modeled as:

$$y_t(n) = A(n) + B(n)^\top F_t + \varepsilon_t(n) \tag{1}$$

where $y_t(n)$ is the observed yield at time $t$ for maturity $n$, $A(n) \in \mathbb{R}$ is a maturity-specific intercept, $B(n) \in \mathbb{R}^k$ is a vector of maturity-specific factor loadings, $F_t \in \mathbb{R}^k$ is a vector of pricing factors, and $\varepsilon_t(n)$ is an error term assumed to be independently and identically distributed (i.i.d.) with zero mean and constant variance. In the affine no-arbitrage framework, the coefficients $A(n)$ and $B(n)$ are not estimated independently for each maturity. Instead, they are linked across maturities through the exponential-affine bond-pricing equations implied by the no-arbitrage restrictions of the model. Consequently, $A(n)$ and $B(n)$ are determined by the risk-neutral dynamics of the factors.

Following Adrian, Crump, and Moench (2013), the pricing factors are obtained from the cross-section of observed yields and are assumed to follow a first-order vector autoregressive process (VAR(1)):

$$F_{t+1} = \Phi F_t + u_{t+1}, \quad u_{t+1} \sim \mathcal{N}(0, \Sigma) \tag{2}$$

The ACM estimation procedure proceeds in two stages. In the first stage, the factor structure is extracted from the cross-section of yields using principal component analysis (PCA). The resulting principal components are used as pricing factors. In the second stage, the market prices of risk are estimated through cross-sectional regressions that relate bond yields to the estimated factors $F_t$ and their conditional expectations, $\mathbb{E}_t[F_{t+1}]$. These conditional expectations are obtained from the VAR representation of the factor dynamics:

$$\mathbb{E}_t[F_{t+1}] = \Phi F_t \tag{3}$$

Using these estimates, the term premium for a bond with maturity $n$ is defined as the difference between the actual yield and the corresponding risk-neutral yield:

$$\mathrm{TP}_t(n) = y_t(n) - y_t^{\mathrm{RN}}(n) \tag{4}$$

The risk-neutral yield $y_t^{\mathrm{RN}}(n)$ is calculated based on expected future factors but excludes compensation for risk, reflecting the yield that would prevail in a hypothetical risk-neutral world.

## 1.2 Principal Components and Factor Models

In the ACM model, the latent pricing factors are estimated using principal component analysis (PCA) applied to a panel of zero-coupon bond yields across maturities. As shown by Stock and Watson (2002), principal components can consistently estimate the space spanned by latent factors under mild regularity conditions, especially when the cross-sectional dimension $m$ grows large relative to the time dimension $T$. A general factor model can be formulated as:

$$Y = F\Lambda + W, \tag{5}$$

where $F \in \mathbb{R}^{T\times k}$ contains the first $k$ principal components (scores), $\Lambda \in \mathbb{R}^{k\times m}$ contains the corresponding loading vectors (eigenvectors), $k \ll m$ is the number of retained components, $W$ is an error vector capturing the approximation error.

Unlike general factor models, PCA assumes orthogonality of components and does not explicitly model the noise structure $W$ but offers a nonparametric alternative to traditional factor estimation, avoiding strong distributional assumptions. However, a key limitation of PCA is the non-uniqueness of its components. While the principal component space is uniquely defined, the individual components are only identified up to an orthogonal rotation. That is, if $F$ denotes the estimated factors, then any rotated version, $\hat{F} = FQ$ , where $Q$ is an orthogonal matrix [1], spans the same space and yields identical fit. This rotational indeterminacy implies that interpretability of components depends on post-estimation choices. Such transformations preserve the statistical properties of the estimated factors while aligning them with theoretical or empirical priors. In summary, while PCA and factor models are tightly linked in their ability to recover latent structures, the interpretability of principal components hinges on rotational choices, with methods like Givens rotations offering a principled path toward meaningful representation.

### 1.3 Reconfiguring Factor Loadings via Givens Rotations

We can alter the interpretation of PCA factors while preserving their statistical properties by a method introduced by Givens (1958), in which a *orthonormal matrix* is used to perform a rotation in the plane spanned by two coordinate axes, typically denoted $i$ and $j$, in $\mathbb{R}^k$. The corresponding *Givens rotation matrix*, $G(i, j, \theta)$, is derived from the identity matrix by modifying the entries in rows and columns $i$ and $j$ to implement the rotation:

$$G(i,j,\theta) = \begin{pmatrix} 1 & 0 & \cdots & & & & & \cdots & 0 \\ 0 & 1 & \cdots & & & & & \cdots & 0 \\ \vdots & \vdots & \ddots & & & & & & \vdots \\ & & & \cos(\theta) & \cdots & \sin(\theta) & & & \\ & & & \vdots & \ddots & \vdots & & & \\ & & & -\sin(\theta) & \cdots & \cos(\theta) & & & \\ \vdots & \vdots & & & & & \ddots & & \vdots \\ 0 & 0 & \cdots & & & & & 1 & 0 \\ 0 & 0 & \cdots & & & & & 0 & 1 \end{pmatrix} \tag{6}$$

The Givens rotation modifies exclusively the $i$-th and $j$-th rows and columns, while all remaining diagonal entries are equal to 1 and all off-diagonal entries are zero. The rotation matrix $G(i, j, \theta)$, where $\theta$ denotes a rotation angle measured in *radians*, mixes columns $i$ and $j$ of $\Lambda$, and simultaneously rows $i$ and $j$ of $F$, preserving orthogonality and vector norms in both matrices. Specifically, the rotated factor matrix is given by a unique transformation based on Givens rotations:

$$\hat{F} = FG(\theta_k)^\top G(\theta_{k-1})^\top \cdots G(\theta_1)^\top \tag{7}$$

$$\hat{\Lambda} = G(\theta_1)G(\theta_2) \cdots G(\theta_k)\Lambda \tag{8}$$

so that the rotated model

$$Y = \hat{F}\hat{\Lambda} + W \tag{9}$$

remains equivalent to the original. These rotations affect only the interpretation of the factors and their loadings, not the statistical content of the model. Gürkaynak, Sack, and Swanson (2005) apply a rotation technique to disentangle monetary policy surprises—measured by

[1] An orthogonal matrix has the properties $Q^\top Q = QQ^\top = I$

changes in federal funds futures across different maturities—into a target factor and a path factor.

### 1.4 Sparse Representation of Yield Curve Loadings through Givens-Based Factor Rotation

Let $\Lambda \in \mathbb{R}^{k \times m}$ denote the loading matrix obtained from PCA, where $m$ is the number of maturities and $k$ is the number of principal components retained. Let $F \in \mathbb{R}^{T \times k}$ be the corresponding time series of factors. We apply a sequence of unique Givens rotations to $\Lambda$ such that the rotated loading matrix $\hat{\Lambda}$ satisfies the following sparsity condition:

$$\hat{\lambda}_{i1} = 0 \quad \forall i > 1 \tag{10}$$

That is, only the first rotated factor contributes to the short end of the yield curve (typically the shortest maturity), while the remaining factors are constructed to be orthogonal to it in terms of their loading structure. This is accomplished by applying a sequence of Givens rotation matrices, where each rotation $G(i, j, \theta)$ is chosen to minimize the magnitude of $\Lambda_{i1}$. In this way, the desired sparsity is enforced in the first row of the loading matrix.

Since only $k-1$ restrictions are required, we construct a sequence of $k-1$ Givens rotation matrices, denoted $G(\theta_1), G(\theta_2), \ldots, G(\theta_{(k-1)})$. The procedure is iterative, and for each rotation $s = 1, \ldots, k-1$, the rotation angle $\theta_s$ is chosen to solve the following:

$$\theta_s = \arg\min_{\theta} |(G(\theta_s)\lambda_{i1})| \tag{11}$$

This optimization can be performed analytically or numerically, depending on the structure of $\Lambda$. In the case where $k = 2$, the rotation angle can be calculated as $\theta^* = \operatorname{arccot}\left(\frac{\lambda_{11}}{\lambda_{21}}\right)$, where $\lambda_{11}$ and $\lambda_{21}$ are the elements of the first column in $\Lambda$. This replicates the solution in Gürkaynak, Sack, and Swanson (2005).[2] In the case where $k > 2$, it is more convenient to rely on a numerical solution. After applying the necessary rotations, the column of $\hat{\Lambda}$ will contain near-zero values in all rows except the first, thereby satisfying the imposed sparsity constraint.

Figure 1 presents the factor loadings across maturities for the first four extracted factors. Each subplot depicts the loading profile over the maturity spectrum for a single factor, where the blue and red lines correspond to loadings obtained via Principal Component Analysis (PCA) and Givens rotation, respectively. The top-left panel (1st factor) clearly illustrates the impact of the rotation: the red line indicates that the first rotated factor has been aligned to load strongly on the shortest maturity, while its loadings approach zero for longer maturities. This suggests that the rotation has successfully isolated a short-term component within the yield curve. In contrast, the remaining three panels show that the red lines—representing the rotated factors—exhibit loadings that are effectively zero at the shortest maturity. This implies that these factors primarily capture variation associated with medium- and long-term maturities. The rotation thus facilitates a disentanglement of the temporal structure embedded in the latent factors.

This sparsity pattern resembles the objective of the varimax (variance maximization) rotation method, originally introduced by Kaiser (1958). Varimax is a widely used orthogonal rotation technique in factor analysis, designed to enhance interpretability by maximizing the variance of squared loadings within each factor. By promoting a "simple structure," varimax redistributes the explanatory power across factors, thereby clarifying the underlying latent constructs.

[2] This follows from the fact that $\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$, and that the restriction is $\cos(\theta)\lambda_{21} - \sin(\theta)\lambda_{11} = 0$.

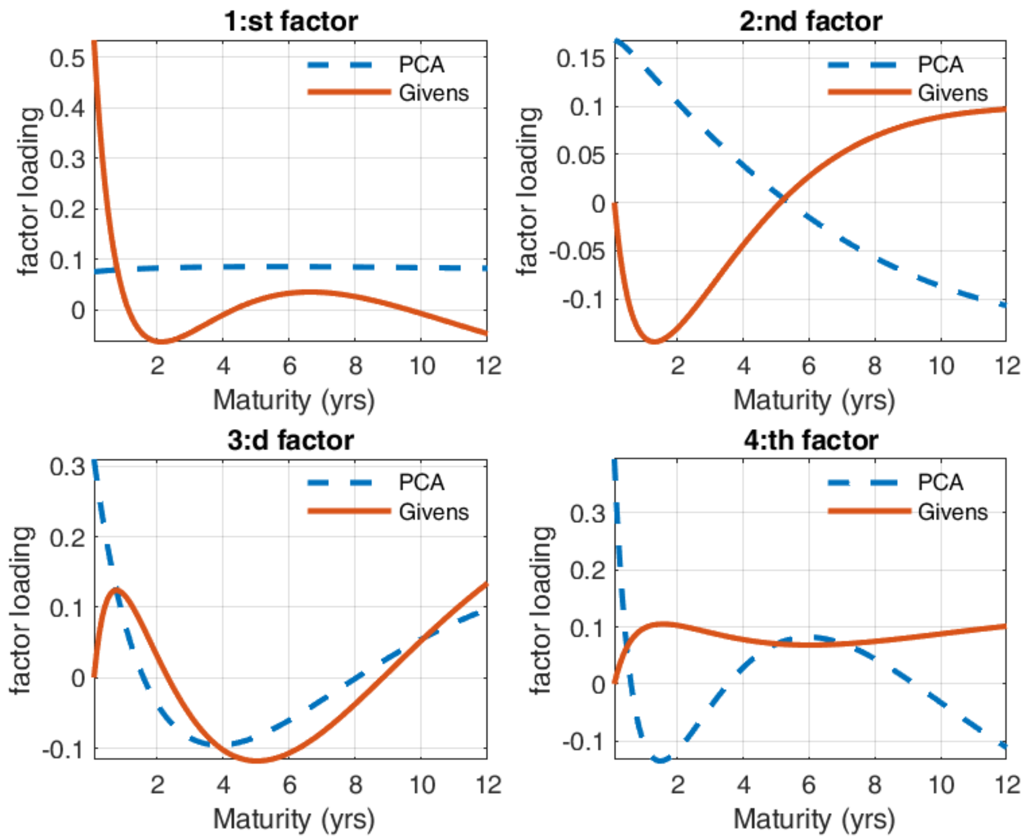


Figure 1: Factor Loadings Across Maturity for PCA and Givens-Rotated Factors, 4 Factor Specification

### 1.5 Block-Orthonormal Rotations in $\mathbb{R}^k$

The sparsity restriction imposed on the first factor ensures that the rotated loading matrix $\hat{\Lambda}$ isolates a short-term component of the yield curve. As illustrated in Figure 1, the first rotated factor loads strongly on the shortest maturity, while the remaining factors exhibit loadings that are identically zero at the short end. This pattern reflects the effect of the initial sequence of Givens rotations: by eliminating all entries $\hat{\lambda}_{i1}$ for $i > 1$, the first factor becomes anchored at the short end of the maturity spectrum, and the remaining factors are orthogonal to it in their loading structure.

Having imposed this restriction, any further rotation of the remaining factors must preserve the structure of the first column. In other words, subsequent transformations must leave the short-end factor unchanged while allowing additional economically meaningful restrictions to be introduced for factors $2, \ldots, k$. This motivates the use of *block-orthonormal rotations*, which operate exclusively on the subspace spanned by the remaining factors and therefore maintain the sparsity pattern established in the first step. After imposing the sparsity restriction on the first factor, namely

$$\hat{\lambda}_{i1} = 0 \quad \forall i > 1,$$

the rotated loading matrix takes the form

$$\hat{\Lambda} = \begin{pmatrix} \hat{\lambda}_{11} & \hat{\lambda}_{12} & \cdots & \hat{\lambda}_{1k} \\ 0 & \hat{\lambda}_{22} & \cdots & \hat{\lambda}_{2k} \\ 0 & \hat{\lambda}_{32} & \cdots & \hat{\lambda}_{3k} \\ \vdots & \vdots & \ddots & \vdots \end{pmatrix}.$$

To preserve this structure while introducing additional economically motivated restrictions on the remaining factors, subsequent rotations must act only on the subspace spanned by factors $2, \ldots, k$. Formally, we consider orthonormal transformations $Q \in O(k)$ of the block-diagonal form

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & Q_{(k-1)\times(k-1)} \end{pmatrix},$$

where $Q_{(k-1)\times(k-1)} \in O(k-1)$ is an arbitrary orthonormal matrix operating on the reduced factor space. The transformation of the loading matrix is then

$$\tilde{\Lambda} = Q\,\hat{\Lambda}.$$

Because the upper-left entry of $Q$ is fixed at 1 and the remaining entries in the first row and column are zero, the first column of $\hat{\Lambda}$ is left invariant:

$$\tilde{\lambda}_{i1} = \hat{\lambda}_{i1} \quad \forall i.$$

Thus, the short-end restriction imposed in the first rotation is preserved exactly.

In practice, the block-orthonormal matrix $Q$ is constructed as a product of Givens rotations whose indices are restricted to the set $\{2, \ldots, k\}$. Each rotation $G(i, j, \theta)$ with $i, j \geq 2$ acts only within the lower-right $(k-1)$-dimensional block and therefore leaves the first factor unchanged. This allows additional structure—such as maximizing the loading of a second factor at a long maturity or imposing sparsity on selected rows—to be introduced without disturbing the previously imposed restriction.

### 1.6 Invariant Transformations in Affine Term Structure Models

Joslin, Singleton, and Zhu (2011) introduce the concept of *invariant transformations* in affine term structure models (ATSMs) to address the issue of observational equivalence. An invariant transformation is an affine transformation of the latent factor space that leaves the model-implied yield curve unchanged. More specifically, if the latent factors are transformed according to

$$F_t^* = DF_t + C,$$

where $D$ is an invertible matrix and $C$ is a constant vector, and the model parameters are adjusted consistently, the resulting bond yields remain identical. Consequently, the observable implications of the model are invariant to affine transformations of the latent factors. This property implies that several affine representations of the same ATSM may generate identical yield curves. As a result, the model is not identified without additional restrictions, since different parameterizations can be observationally equivalent. To resolve this indeterminacy and obtain a unique and economically interpretable representation, Joslin, Singleton, and Zhu (2011) propose a canonical normalization that aligns the latent factors with economically meaningful features of the yield curve.

### 1.7 Rescaling the First Factor Using Linear Regression

After applying the sequence of Givens rotations to enforce the constraint that all loadings in the first row, except for the first column, are approximately zero, the first factor is interpreted as a close proxy for the short-term risk-free interest rate. To align the scale of this factor with economic meaning, it is rescaled so that its mean and variance match those of the observed short-term risk-free rate during the sample period. Let $f_1^{\text{rotated}}$ denote the time series of the first factor after rotation, and let $y_{1t}$ denote the observed short-term risk-free rate at time $t$. This scaling can be performed using an ordinary least squares (OLS) regression:

$$y_{1t} = \alpha + \beta f_{1,t}^{\text{rotated}} + \varepsilon_t \tag{12}$$

The estimated coefficients $\hat{\alpha}$ and $\hat{\beta}$ are then used to transform the factor:

$$f_1^{\text{rotated, scaled}} = \hat{\alpha} + \hat{\beta} f_1^{\text{rotated}} \tag{13}$$

This approach ensures that the scaled factor matches the level and volatility of the short-term interest rate, facilitating interpretation and further use in economic modeling. Importantly, this procedure constitutes an affine invariant transformation of the pricing factors, as defined by Joslin, Singleton, and Zhu (2011), preserving the model-implied yields while enabling economically meaningful identification.

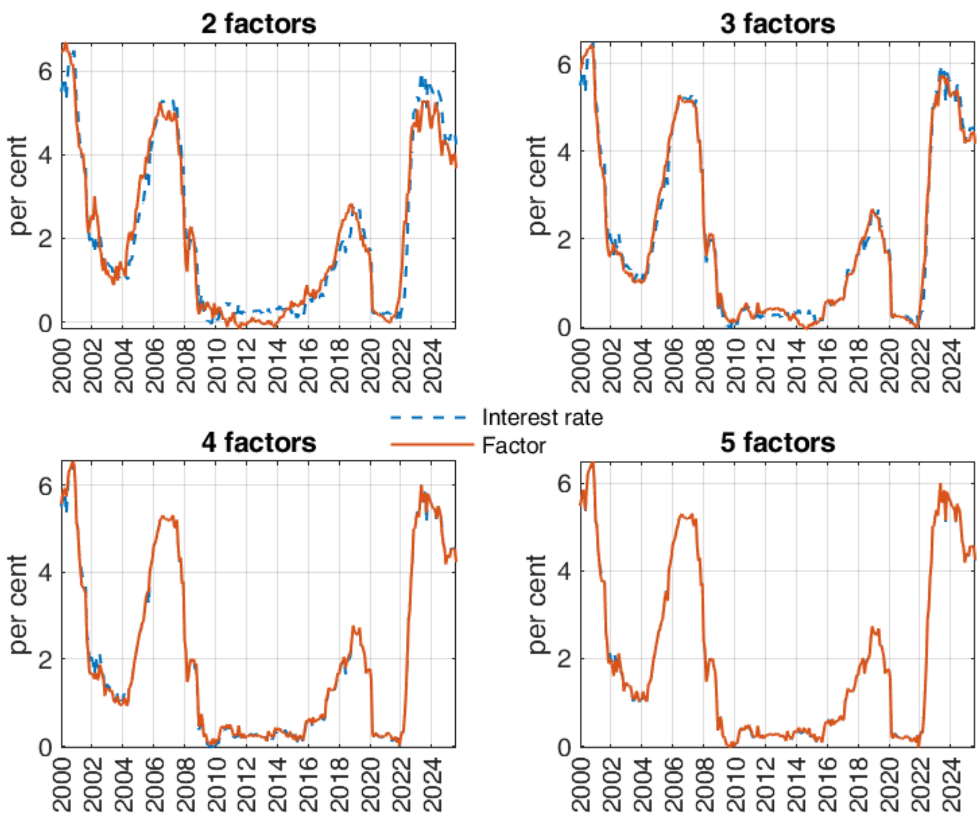


Figure 2: Short-Term Risk-Free Interest Rate and Scaled Givens-Rotated First Factor Across Specifications with two, three, four and five Factors

Figure 2 displays the actual short-term risk-free interest rate (blue line) alongside the first factor extracted from principal component analysis of the yield curve (red line), after applying Givens rotations with imposed sparsity constraint and rescaling to match the level and volatility of the short rate. Each subplot corresponds to a specification with a different total number of factors, ranging from 2 to 5. The results reveal a clear pattern: as the number of factors increases, the rotated and scaled first factor more closely approximates the actual short-term interest rate. This indicates that higher-dimensional factor structures allow for more accurate isolation of the component most closely associated with short-rate dynamics. The improvement is particularly evident when moving from 2 to 4 factors, with marginal gains beyond that point. Notably, the widely used ACM model employs five factors, which appears to offer the best fit among the specifications shown.

# 2 Bayesian VAR Estimation Incorporating Long-Run Mean Beliefs

In line with Meldrum and Roberts-Sklar (2015), long-run priors can be incorporated into the estimation of term structure models to reflect economic theory and historical patterns, particularly regarding equilibrium interest rates and inflation expectations. Such priors can help stabilize the estimation process and mitigate overfitting to short-term noise in the data.

Departing from the approach of Meldrun et al. we employ rotated and scaled factors in our specification. Specifically, our estimation builds on the Bayesian VAR framework introduced by Villani (2009), which enables the integration of prior beliefs about the long-run means of the pricing factors. To model the term structure of interest rates, we adopt the flexible and empirically grounded framework developed by Adrian, Crump, and Moench (2013), which allows for a rich representation of yield curve dynamics. Importantly, our use of Givens rotations facilitates the identification of economically interpretable factors, and the scaling procedure aligns these factors with observable short-term interest rates. As demonstrated in our results, increasing the number of factors improves the alignment between the rotated first factor and the actual short rate, an observation consistent with the five-factor specification employed in the ACM model. Let

$$F_t^* = (f_{1,t}^{\text{rotated, scaled}}, f_{2,t}^{\text{rotated}}, \ldots, f_{n,t}^{\text{rotated}})^\top$$

denote the vector of rotated and transformed latent pricing factors extracted from the yield curve. The BVAR(1) model is specified as:

$$F_t^* = \mu + \Phi(F_{t-1}^* - \mu) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \Sigma) \tag{14}$$

where $\mu$ is the vector of unconditional means, $\Phi$ is the matrix of autoregressive coefficients, and $\Sigma$ is the innovation covariance matrix. This mean-centered formulation ensures that the dynamics of the factors is anchored around economically interpretable long-run levels.

## 2.1 Incorporating Long-Run Beliefs via Hierarchical Priors

Villani (2009) proposes a hierarchical prior structure that explicitly targets the unconditional mean $\mu$ of the VAR process. He introduces a hierarchical prior structure that allows the unconditional mean vector, $\mu$, to be treated as a parameter with its own prior distribution. This enables researchers to incorporate economically meaningful beliefs about long-run levels—such as the steady-state interest rate—into the model. The hierarchical setup improves flexibility, especially in applications such as yield curve modeling, where anchoring certain factors to plausible long-run values is essential. The prior on $\mu$ is specified as:

$$\mu \sim \mathcal{N}(\mu_0, V_\mu) \tag{15}$$

where $\mu_0$ is the prior mean vector and $V_\mu$ is the prior covariance matrix. This allows the researcher to encode beliefs about the long-run (steady-state) values of the system. The posterior distribution of $\mu$ is updated jointly with the VAR parameters using Gibbs sampling. In the context of yield curve modeling, this is particularly useful for anchoring the short-rate factor to a plausible long-run level, while maintaining flexibility for the remaining factors. In this application, the prior mean of the first factor is set to the subjective belief about the long-term mean of the short-term interest rate, reflecting the expectations of the monetary policy. The prior means of the remaining factors are set to zero, consistent with their interpretation as the mean-zero principal components derived from the yield curve.

### 2.2 Incorporating Survey Expectations through Predictive Density Restrictions

An alternative way of incorporating information about the expected path of short-term interest rates is to use survey expectations rather than direct priors on the unconditional mean of the factors. This approach is closely related to the methodology of Kim and Orphanides (2005), who augment affine term structure models with survey forecasts of future short rates. Whereas the steady-state prior of Villani (2009) primarily influences the unconditional mean of the VAR, survey expectations provide information about the expected transition dynamics of the factors over specific forecast horizons. With the rotated-factor specification employed here, the first factor is constructed to closely track the short-term interest rate. Let the first element of $F_t^*$ be denoted by $f_{1,t}$. Under the VAR(1) dynamics

$$F_t^* = \mu + \Phi(F_{t-1}^* - \mu) + \varepsilon_t, \tag{16}$$

the model-implied expectation of the short rate at horizon $h$ is

$$m_t(h) = e_1' \left[ (I - \Phi^h)\mu + \Phi^h F_t^* \right], \tag{17}$$

where $e_1$ is a selection vector with unity in the first position and zeros elsewhere. Suppose survey forecasts are available for horizons $h_1, \ldots, h_K$. Let

$$s_t = \begin{bmatrix} s_t(h_1) \\ \vdots \\ s_t(h_K) \end{bmatrix} \tag{18}$$

denote the corresponding survey observations. Following the spirit of Kim and Orphanides (2005), the discrepancy between survey expectations and model-implied forecasts can be interpreted as a forecast error,

$$u_t = s_t - m_t, \tag{19}$$

where $m_t$ is the vector of model-implied expectations at the same horizons. The survey forecasts are then treated as noisy observations of the model-consistent conditional expectations,

$$u_t \sim \mathcal{N}(0, \Omega), \tag{20}$$

with covariance matrix $\Omega$ governing the degree of confidence placed on survey information. Conditional on the VAR parameters, the contribution of the surveys to the likelihood function is therefore

$$L_{\text{survey}} \propto |\Omega|^{-T_s/2} \exp\left( -\frac{1}{2} \sum_{t=1}^{T_s} u_t' \Omega^{-1} u_t \right), \tag{21}$$

where $T_s$ denotes the number of survey observations.

In contrast to the Gaussian conditional posteriors obtained in the original Gibbs sampler of Villani (2009), the inclusion of survey information introduces nonlinear dependence on the VAR transition matrix $\Phi$ through the terms $\Phi^h$. As a consequence, the conditional posterior distribution of $(\Phi, \mu)$ is no longer available in closed form. The Gibbs sampler must therefore be augmented with a Metropolis-Hastings step. Specifically, after drawing the covariance matrix $\Sigma$ and the remaining parameters from their standard conditional posterior distributions, a candidate draw $(\mu^*, \Phi^*)$ is generated from a proposal density. The acceptance probability is

$$\alpha = \min\left(1, \frac{p(Y|\mu^*, \Phi^*, \Sigma)L_{\text{survey}}(\mu^*, \Phi^*)p(\mu^*, \Phi^*)}{p(Y|\mu, \Phi, \Sigma)L_{\text{survey}}(\mu, \Phi)p(\mu, \Phi)}\right), \tag{22}$$

where $p(Y|\mu, \Phi, \Sigma)$ denotes the standard VAR likelihood and $p(\mu, \Phi)$ represents the prior distribution. The survey likelihood therefore acts as an additional weighting mechanism, favoring parameter configurations that generate forecasts consistent with observed survey expectations.

This approach can be viewed as a dynamic alternative to imposing informative steady-state priors. Whereas long-run mean priors primarily anchor the asymptotic behavior of the system, survey expectations provide information about the expected path of future short rates at specific horizons. Consequently, survey information tends to identify the persistence parameters in $\Phi$, while long-run priors mainly identify the unconditional mean $\mu$. Since both mechanisms contain information about future rates, they may interact strongly in estimation and can potentially give rise to identification issues when overly informative priors and survey restrictions are imposed simultaneously. In practice, we find that survey expectations and long-run mean priors provide overlapping information about the future evolution of short rates. Consequently, careful calibration of the survey error covariance matrix $\Omega$ and the prior covariance matrix $V_\mu$ is required in order to avoid excessive identification of the same underlying dynamics.

## 2.3 Yield Curve Representation and Implications

The estimated factors $F_t^*$ are subsequently used in the representation of the affine yield curve:

$$y_t(\tau) = A(n) + B(n)^\top F_t^* + \eta_t(n) \tag{23}$$

where $y_t(n)$ denotes the yield at time $t$ with maturity $n$, and $A(n)$ and $B(n)$ are maturity-specific intercept and loading vectors. The error term $\eta_t(n)$ captures the measurement noise.

This modeling approach is consistent with the empirical strategy employed in the ACM framework and offers a coherent structure for extracting and forecasting latent yield curve factors while incorporating long-term economic beliefs. As emphasized by Meldrum and Roberts-Sklar (2015), the use of informative priors on unconditional means in affine term structure models can enhance the stability of the estimated dynamics.

Figure 3 illustrates how the model's estimate of the term premium is shaped by the choice of prior in the framework of Villani (2009). This effect arises because the prior influences the model's forecasts of the short-term interest rate, which in turn feed directly into the calculation of the term premium. When a prior is used that centers around the sample mean of the short rate since 1994—a period characterized by relatively low interest rates—the estimated term premium is notably higher. Conversely, employing a prior centered around the sample mean since 1961, which includes periods of higher average rates, results in a lower estimated term premium. A prior that lies between these two extremes yields an intermediate estimate of the term premium. These results highlight the sensitivity of term structure estimates to prior assumptions, underscoring the importance of carefully selecting priors in empirical applications.

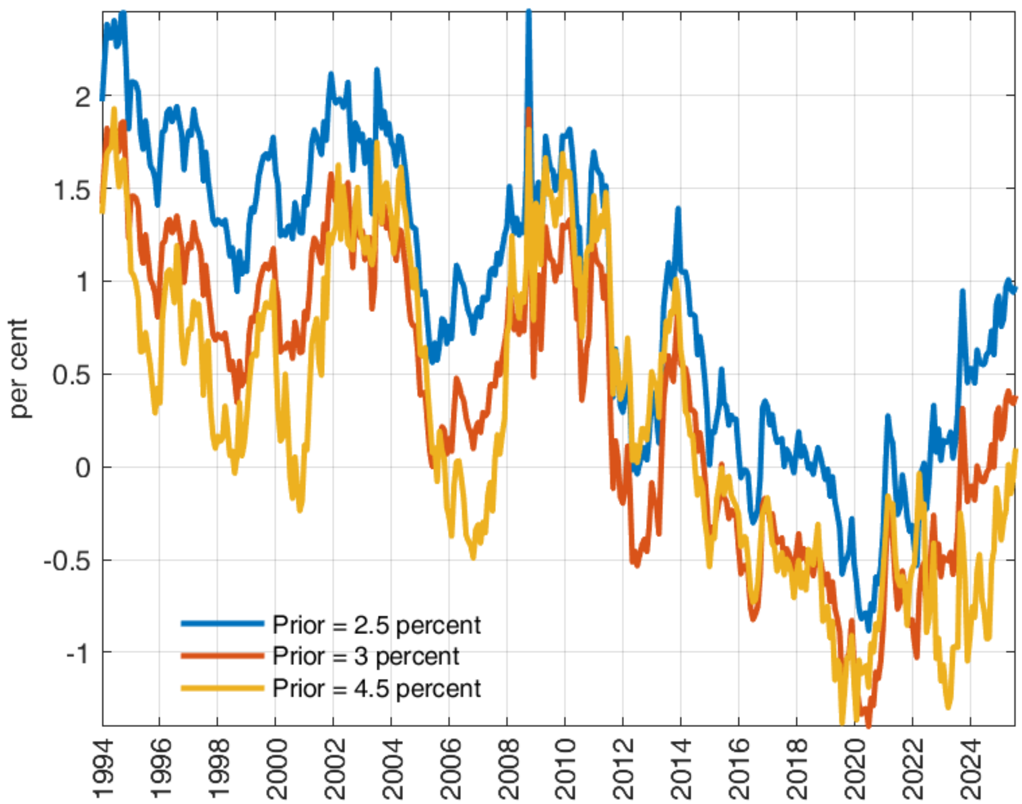


Figure 3: Estimated 5-Year US Term Premium Using ACM and Alternative Priors for the Short-Term Interest Rate. The estimates are based on monthly data from January 1994 to August 2025, using maturities of 1 and 6 months, and 1 to 11 years.

## 3 Discussion

The proposed modification of the ACM framework preserves the statistical efficiency of principal component analysis (PCA) while introducing an economically meaningful structure through rotation, scaling, and Bayesian estimation. By imposing sparsity in the loading matrix and aligning the first factor with the short-term interest rate, the model improves the interpretability of yield curve dynamics and facilitates the integration of macroeconomic priors into term structure modeling.

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